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The Sociology of Innovation: Organizational, Environmental, and Relative Perspectives

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Abstract

Innovation is risky. New products and firms are subject to high failure rates. To minimize the risk associated with innovation, most scholars agree that firms should engage simultaneously in two types of activities: exploring new alternatives and exploiting existing competencies. Firms that simultaneously engage in explorative and exploitative activities are called ambidextrous organizations. Research on ambidextrous organizations, however, inadequately considers the importance of the environment in which organizations operate for innovation success. The sociological view of innovation fills this gap by emphasizing the importance of social context for explaining innovative outcomes. The sociology of innovation highlights, first, the structural arrangements – characteristics of social networks, organizations, and institutions – that influence innovation and, second, the relative nature of innovation – whether an object is considered novel depends on one's vantage point. Drawing on the relative view of innovation, I develop a typology that outlines several paths to innovation with respect to two vantage points: the organization and the environment. I argue that minimizing the risks associated with innovation is most likely to occur when firms follow the path of innovation that consists of generating novel solutions that simultaneously exploit knowledge and resources that are both available to the firm *and* in the firm's environment.

Introduction

Despite organizations' best efforts to adapt in today's ever-changing, global economy, attempts to innovate often fail. Many new firms fail within the first years of operation and most disappear in subsequent years. One study found that 10 percent of US businesses survive after 10 years and about one-tenth of 1 percent survive after 40 years (Stubbart and Knight 2006). Most new products and services fail as well (Pfeffer and Sutton 2006) and even the most successful products can take years, if not decades, to diffuse throughout society (Rogers 1995). In the pharmaceutical industry, the Pharmaceutical Research and Manufacturers of America (2007) estimates that a chemical compound developed by any given firm has a 0.02 percent chance of eventually evolving into a drug that goes to market. Overall estimates of new product success suggest between 30 to 60 percent of new products sold on the market fail to generate a profit, though profit is just one metric for evaluating product success (Schilling and Hill 1998).

The pressure to adapt in today's rapidly changing environment, on the one hand, and the risk of failure resulting from change, on the other hand, represent a fundamental tension for organizations. Haveman (1992, p. 51) puts it this way: "There is an inherent paradox here. If change increases the risk of failure, why do organizations ever try to change?" Most business management scholars agree the answer to the question about resolving this innovation paradox is that firms should engage in two types of activities simultaneously: exploration – exploring new alternatives – and exploitation – exploiting existing competencies (Abernathy 1978; Benner and Tushman 2003; Duncan 1976; Gibson and Birkinshaw 2004; Gupta et al.

2006; Levinthal and March 1993; March 1991; O'Reilly and Tushman 2008). Exploration permits firms to adapt to changes in the environment and capture new customers (Tushman and Smith 2002) thereby increasing the likelihood of long-term success while exploitation enhances the prospects of short-term performance (Levinthal and March 1993; March 1991, 2003).¹ Firms that undertake exploration and exploitation activities simultaneously are referred to as ambidextrous organizations (for reviews, see Lavie et al. 2010; O'Reilly and Tushman 2008; Raisch and Birkinshaw 2008).

Research on organizational ambidexterity, however, would benefit by paying more attention to the environment in which organizations operate. Firms may explore new alternatives and exploit existing resources or competencies that exist in the organizational environment in addition to engaging in exploration and exploitation activities within the boundaries of the firm. The organizational environment consists of "all those significant elements outside the organization that influence its ability to survive and achieve its ends" (Scott and Davis 2007, p. 19). Elements in the environment include other actors such as "the clients, constituents, or customers that the organization serves and the producers of resources it requires to do so" (Scott and Davis 2007, p. 20). The "clients and constituents" referenced by Scott and Davis may include the state, strategic alliance partnerships between firms, competitors, and social movements.

Organizational scholars have long recognized the role that environmental conditions play in organizational outcomes (Stinchcombe 1965; Thompson 1967). To be successful, organizations depend on resources available in the environment (Pfeffer and Salancik 1978) and legitimacy (Hannan and Freeman 1989; Meyer and Rowan 1977) conferred on organizations that conform to government regulations, professional standards, and best practices of other successful firms (DiMaggio and Powell 1983). Organizations also learn from the experience of others (Levitt and March 1988) and benefit by pooling their expertise and resources (Powell et al. 1996; Uzzi 1997; Uzzi and Lancaster 2003; Van de Ven 2005). Firms increasingly employ "open innovation" strategies by looking outside their boundaries to acquire new products and technology (Chesbrough 2003). Actors in the environment such as strategic alliance partners (Hagedoorn and Schakenraad 1994; Powell et al. 1996; Shan et al. 1994; Stuart 2000; Uzzi 1997) and end-users (Bogers et al. 2010; Boudreau and Lakhani 2013; von Hippel 2005) often prove to be valuable sources of new ideas.

The burgeoning literature on the sociological view of innovation emphasizes the environment in explanations of successful innovation. While it is impossible to recount all of the literature on the sociological factors that impact innovation success, this paper provides an overview of key studies and concepts that support the sociological view of innovation. The sociology of innovation highlights the structural arrangements – characteristics of social networks, organizations, and institutions – within the social context that influence innovation. As well, this view emphasizes the relative or subjective nature of innovation. That is, whether a product is considered novel is subjective because it depends on the one's point of view. A new product might be radically different from products developed by a particular firm (exploration), but the same product or a slight variation of the product may already exist in the environment (exploitation). Alternatively, a product may be viewed as radically new from the point of view of actors in the environment (exploration), but represent a minor change or no change at all from a particular firm's products (exploitation). In (Dahlin 2011), I describe an interview I conducted with the CEO of a small pharmaceutical firm that illustrates the subjective nature of innovation. The CEO began the interview by stating that his firm's products are extremely innovative. His firm focused on "next generation" drug candidates that represented a radical departure from the chemical compounds used by other firms in the industry. Nevertheless, further discussion revealed that each subsequent drug this firm developed and tested in clinical trials was based on the same technology and,

consequently, represented only a slight modification to drugs already being developed by the firm. Initially, the product development strategy described by the CEO was innovative – but only with respect to *other firms'* drugs. Viewed from the vantage point of *this firm*, each new drug represented only a minor change from the drugs already developed.

Not all vantage points are equal, however. Some points of view are more insightful or provide greater advantage than others. Which views are most consequential and likely to contribute to innovation success? Drawing on the notion that different vantage points influence whether a product is deemed novel, I develop a typology that outlines paths to innovation that consist of various combinations of exploration or exploitation with respect to two vantage points: the organization and the environment. In order to reduce the challenges associated with innovation, I argue that the most promising path to innovation success organizations can follow consists of generating novel products that exploit both firm competencies *and* knowledge and resources available in the environment.

The interface between exploration and exploitation

The organizational learning perspective focuses on the way organizations produce knowledge based on past experience. Organizational learning processes encode “inferences from history into routines that guide behavior” (Levitt and March 1988, p. 320). Routines become part of the informal and formal structure as organizations learn, informally through social norms and formally through manuals, goals, and reports. Encoded rules preserve prior organizational experience and provide cues for reducing uncertainty (Levitt and March 1988; March and Simon 1993; Nelson and Winter 1982). To the extent that prior experience is encoded in the organizational structure through rules and procedures, it shapes future activities by reducing variation in beliefs, behaviors, and performance and mediates potential changes due to exogenous shocks in the organization's environment (Zhou 1993).

Although these processes can limit innovation by decreasing the range of alternatives organizations are likely to pursue (Levinthal 1991; Zhou 1993), organizational learning approaches to innovation also concentrate on routines designed to search for and generate new knowledge (Argote and Miron-Spektor 2011). Search routines exist for the purpose of finding new solutions, developing better ways to perform tasks, and combining extant knowledge in new ways (Cyert and March 1963; Nelson and Winter 1982). March (1991) identifies two learning activities, exploration and exploitation, that organizations utilize to generate varying degrees of novelty. Exploration is the “pursuit of new knowledge” (Levinthal and March 1993, p. 105) and involves radical deviations from existing practice. Exploration allows firms to capture new customers and adapt to new developments in the market. Consequently, exploration increases the likelihood of a firm's long-term success (Levinthal and March 1993; March 1991, 2003), and it is an especially important strategy for firms operating in rapidly changing environments. In contrast, exploitation, as noted by March (1991, p. 85), “is the refinement and extension of existing competencies, technologies, and paradigms.” Exploitative learning is incremental, based on minor improvements to existing products or practices. Exploitation leads to more stable and quicker outcomes because organizational routines often improve as they repeat the same or similar tasks. Compared with exploration, the benefits of exploitation include increased efficiency and decreased production costs. Exploitation enhances short-term performance (Levinthal and March 1993; March 1991, 2003), and it is a beneficial strategy in (stable) environments characterized by minimal change.

Firms that pursue exploration and exploitation activities simultaneously are referred to as ambidextrous organizations (Duncan 1976; Gibson and Birkinshaw 2004; Gupta et al. 2006; He

and Wong 2004; Lavie et al. 2010; O'Reilly and Tushman 2008; O'Reilly and Tushman 2008; Raisch and Birkinshaw 2008; Raisch et al. 2009). Ambidextrous organizations are more likely to realize the benefits associated with pursuing both explorative and exploitative actions. Firms that engage in some degree of exploration and exploitation are thought to be more profitable because they are better able to adapt in changing environments while leveraging their competencies.

Still, current conceptions of organizational ambidexterity could benefit from paying closer attention to the organizational environment. First, exploration and exploitation activities can occur beyond the boundaries of the firm, and research that uses an ambidexterity perspective to examine the ways firms acquire novel (exploration) or familiar (exploitation) knowledge and resources from external sources is scarce. Scott and Davis (2007, pp. 19–20), among others (c.f. Pfeffer and Salancik 1978), highlight the importance of the environment for providing resources for organizational activities when they state, “The environment can be seen as a store of resources as well as a source of opportunities and constraints, demands and threats.” Second, the ambidexterity literature, and much of the research on innovation, often overlooks the notion that a product can be considered novel or familiar depending on one's point of view. The perspective that novelty depends on one's point of view is informed by the relative view of innovation, which is discussed in greater detail below.

Firms are aided by external sources in the innovation process through feedback from customers and end-users who often custom-design and modify many products to fit their individual needs (Bogers et al. 2010; Boudreau and Lakhani 2013; von Hippel 2005) as well as from relationships with other firms (Powell et al. 1996; Uzzi 1997; Uzzi and Lancaster 2003; Van de Ven 2005). Research indicates that entering relationships with other firms – which can take a variety of forms including temporary contracts or licensing agreements, joint ventures or strategic alliance partnerships organized around the codevelopment of products or services and the sharing of information, and mergers and acquisitions (Todeva and Knoke 2005) – fosters the sharing of knowledge and expertise that can increase the rate of learning and innovation (Hagedoorn and Schakenraad 1994; Shan et al. 1994; Stuart 2000; Uzzi 1997).

The sociology of innovation fills these gaps in current approaches to organizational ambidexterity and sheds additional light on the innovation paradox. The remainder of this paper outlines the sociology of innovation and concludes by discussing the implications for the innovation paradox. In the following section, I provide an overview of the sociology of innovation by highlighting the social network, organizational, and institutional arrangements that affect innovation as well as the relative nature of innovation. Then, I elaborate several paths to innovation that draw on insights that follow from viewing novelty as a subjective phenomenon.

The sociology of innovation

While popular accounts have historically attributed innovation success to insightful inventors or the form and function of product design, the sociology of innovation emphasizes the importance of social context for understanding the development and success of novel products and services. According to the sociological view, social arrangements shape the innovation process and its outcomes (Daniel and Dawson 2011; MacKenzie and Wajcman 1999). For instance, Becker's (1982) research illuminates how the production of art is the result of collective effort that includes formally trained artists, those who supply materials and maintain instruments used by artists, and the critics and audiences who define its ultimate success. Not to mention the cooperative behavior of actors in these social networks is fostered and constrained by social conventions that prescribe interactions between relevant constituents,

appropriate techniques for creating art or music and the materials that are used, and organizational arrangements necessary to support artistic endeavors.

Social network, organizational, and institutional arrangements

While much sociological research on the relationship between social networks and innovation has been concerned with the diffusion of innovation (Strang and Soule 1998), social ties facilitate the sharing of knowledge and expertise that generate new ideas within organizations (Burt 2002; Hargadon and Sutton 1997; Obstfeld 2005) and between organizations (Hagedoorn and Schakenraad 1994; Powell et al. 1996; Shan et al. 1994; Stuart 2000; Uzzi 1997; Van de Ven 2005). Actors who occupy strategic positions in networks are more likely to have access to diverse sources of knowledge and resources and to recombine them in new ways. Network structures that have been shown to foster novelty include structural holes, small world networks, and structural folds.

Burt (1992) develops the notion of structural holes to refer to an actor who occupies a position of advantage in a social network. According to Burt, an actor who bridges a gap between others who are unconnected is considered a broker who spans a structural hole. Actors in this position who bridge gaps between unconnected others are more likely to have access to diverse sources of knowledge and provide non-redundant (i.e. novel) information to others (Hargadon and Sutton 1997; Hillmann and Aven 2011; Podolny and Baron 1997). "With early access to diverse, often contradictory, information and interpretations, people whose networks span structural holes can expect to find themselves synthesizing new understandings.... Putting aside the individuals involved, creativity and learning will occur more often where relationships bridge structural holes" (Burt 2002, p. 158). In a study conducted in a large electronics company, Burt (2004) finds that managers whose networks bridge groups are more likely to generate the best ideas. Additional advantages, including higher salaries, performance evaluations, and promotions, also accrue to people who bridge structural holes (Burt 2005).

Small world networks are characterized by highly connected clusters as well as bridges between clusters that generate relatively short distances between any two people in the entire network (Watts 1999, 2004). Ties to other groups provide access to new and diverse knowledge and allow innovative ideas to spread through the network, while cohesive clusters promote the cooperation and familiarity necessary to implement new ideas (Fleming and Marx 2006; Schilling and Phelps 2007; Uzzi and Spiro 2005). The social psychologist Stanley Milgram is the first to note the characteristics of small world networks (Fleming and Marx 2006; Uzzi et al. 2007). Milgram's (1967) classic experiment was designed to find out how many intermediaries were needed for a letter to be delivered from someone in Omaha, Nebraska to a stockbroker in Boston, Massachusetts. Of the letters that eventually arrived, the average number of steps needed to reach the target was six. More recent research by Uzzi and Spiro (2005) shows that members of teams that produce successful Broadway musicals exhibit features of small world networks. Teams with moderate levels of diversity and cohesion from one musical to the next are more likely to achieve success on Broadway.

Several researchers (de Vaan et al. 2012; Vedres and Stark 2010) build on these strands of research by developing the notion of structural folds. Instead of novel ideas being transferred between previously unconnected individuals and groups, Vedres and Stark underscore the creative capacity of people with overlapping memberships in multiple cohesive groups. Structural folds constitute a position in a network attributed to an actor who occupies a position in multiple groups. Actors in these positions are able to recombine diverse sets of knowledge that are specific to each group to generate novel solutions. The rationale is

described by de Vaan et al. (p. 4) who explain that structural folding generates tensions between the “rules, roles, and codes” of intersecting groups and “promotes successful innovation.” Findings by Vedres and Stark indicate that the presence of structural folds is associated with increasing performance of entrepreneurial groups in Hungary, and de Vaan et al. show that structural folding increases the likelihood that the games created by teams of video game developers are novel and successful.

A variety of organizational arrangements likewise affect innovation. Traditionally, organizational researchers studied the effects of broad organizational characteristics on innovation (e.g. Damanpour 1991). A prominent contribution to this strand of research is a study by Sorensen and Stuart (2000) who point out two ostensibly contradictory effects of organizational aging on innovation: structural inertia and organizational competence. Structural inertia refers to the tendency for organizations’ procedures and practices to withstand change. Investment in operating equipment and personnel (“sunk costs”) generates inertia and routines become taken-for-granted over time. This rigidity of routines ends up limiting the range of appropriate behavior (Hannan and Freeman 1984). Thus, newer organizations are more likely to innovate, but as organizations age, the concomitant effects of structural inertia result in a decreasing number of innovations. An alternative perspective identified by Sorensen and Stuart emphasizes the role of organizational competence in innovation over time. According to this view, younger firms are disadvantaged with respect to innovation. Older firms have more production experience, stronger relationships with customers, more experienced employees (Stinchcombe 1965), and larger reservoirs of accumulated knowledge; all of which can improve firms’ ability to innovate. Sorensen and Stuart resolve the tension between structural inertia and organizational competence by examining the effect of firm age on innovation. In their study of firms in the semi-conductor and biotechnology industries, they find that organizational age has a nonlinear effect. Innovation success increases as younger firms gain more experience, while success levels off and decreases for the oldest firms in their study.

As successful firms age, the size of their operations tends to expand. They increase in the number of employees, operating expenses, and revenue. Therefore, larger firms are more likely to be affected by the forces of structural inertia while smaller firms are generally considered more flexible and better innovators. Among US technology firms with 15 or more patents granted from 2002 to 2006, small firms accounted for more patents per employee and more citations compared with large firms (Breitzman and Hicks 2008). Other research by Block and Keller (2009) demonstrates that the propensity of the largest firms in the United States to produce prize-winning innovations (as determined by *R&D Magazine*) has declined over the past three decades, especially among manufacturing firms. Pisano (2006) discusses the typical model of innovation in the pharmaceutical industry, which also consists of smaller firms that specialize in new drug development. These small, innovation firms develop a new product and then partner with large pharmaceutical companies with the resources to carry out clinical trials for the product, the capacity to manufacture and market the product, and marketing and sales personnel to carry the product to the far corners of the market.

Research on organizational innovation additionally underscores particular practices and routines that contribute to innovation success. Hargadon and Sutton’s (1997) study of IDEO demonstrates how routines for storing, searching for, and retrieving product information with which designers had previous experience served to recombine existing knowledge and generate innovation. Designers at IDEO often accumulated a wide variety of products that were stored in their offices and other centrally located areas. These products proved to be a source of inspiration at a future date when the designers were thinking about how to create

additional, different products. For example, “designers attempting to develop a spill-proof nozzle for a bicycle water bottle described how they recognized similarities between the problems of the water bottle design and a previous project designing a shampoo bottle that could hang upside down” (pp. 738–39). IDEO also deliberately assembled designers with different, but related, areas of expertise to work on projects and share knowledge.

Research on heterarchical organizational forms illuminates practices that promote the recombination of ideas to generate innovation (Beunza and Stark 2003; Girard and Stark 2002; Stark 1996, 2001, 2009). Heterarchy, in contrast with hierarchy, is characterized by distributed decision-making capacity and lateral ties between organizational units. Heterarchical organizations are decentralized and organized around lateral network ties and communication channels that cut across organizational units so that a variety of people participate in decision-making processes. Decision-making that is distributed among a wide range of actors in the organization creates dissonance as actors develop multiple ways to interpret and evaluate circumstances, and multiple and competing paths to problem solving and accomplishing objectives. “Because there is not one best way or single metric but several mutually coevolving yet not converging paths, the organization is systemically unable to take its routines or its knowledge for granted” (Stark 2009, p. 27).

As a result of the friction created by multiple performance criteria, heterarchy promotes perpetual recombination, reorganization, and novelty. Beunza and Stark (2003) illustrate how heterarchy permitted a financial firm to innovate as it resumed operations quickly in the wake of the 9/11 terrorist attack of the World Trade Center in New York City. Due to the attack, the firm was forced to relocate from the World Financial Center, next to the World Trade Center, to a warehouse in neighboring New Jersey and, remarkably, it reopened on September 17. Buena and Stark argue that the heterarchical practices employed by the firm, like diverse social networks, redundant ties between employees, and recombination of old and new technologies, increased employees’ ability to adapt and generate new solutions, since many of the previous operating procedures under the new, post-9/11 conditions were either insufficient or infeasible. Another study by Girard and Stark (2002) documents how a web design firm employed distributed decision-making practices, high levels of communication between organizational units, and simultaneous engineering to keep pace with ever-changing technology, market conditions, and customer demands.

Institutional arrangements impact innovation as well. Institutional arrangements refer to the “governmental agencies, professional trade associations, and scientific/technical communities that legitimate, regulate, and standardize a technology” (Van de Ven 2004, p. 133). The state, for instance, can shape innovative activity by providing funding for or passing laws that support the development of new technologies and markets. Block and Keller’s (2009) research demonstrates the importance of government funding for prize-winning innovations in the United States. In addition, Campbell and Lindberg (1990) explain the state’s role in the evolution of the US commercial nuclear energy industry. As an effort to boost activity in the private sector, in 1954, Congress prohibited the federal government and permitted private firms to build nuclear reactors to produce and sell electricity. “This move helped create a market to coordinate further development and dissemination of [nuclear] technology” (p. 639).

Supportive institutional arrangements influenced the emergence of the synthetic dye industry in Western Europe prior to World War I. Production of synthetic dye began in the latter part of the 1850s in a number of countries including Britain, France, Germany, and Switzerland (Murmah and Homburg 2001). Although British firms were initially the leaders in the industry, Germany quickly became the site of the most successful synthetic dye

producers due its university system, government support of higher education, and strong professional and trade associations. First, academic training in chemistry was critical for learning how to produce synthetic dye and many improvements in products and production processes came from academe. Compared with other countries, chemistry courses and research laboratories proliferated in Germany universities and Germany “possessed the best research and training system in synthetic organic chemistry” (Murmman 2003, p. 69). Moreover, with the expansion of chemistry curriculum and research within academe, available labor pools increased and German firms were able to collaborate extensively with and also hire chemistry faculty and students to work in their corporate R&D labs. Second, public support for education was high in Germany and the government provided a great deal of financial support to higher education institutions; no doubt providing a valuable source of funding for university labs conducting research in the area of synthetic chemistry. For example, in 1904–1905, Britain spent £69,000 on grants to universities. By comparison, Germany’s largest state, Prussia, spent £476,000 a few years earlier in 1900–1901 (Murmman 2003; see also Haber 1971, p. 41). Third, professional and trade associations in chemistry were more influential in Germany than elsewhere. Professional associations distributed knowledge to chemists through meetings and publications (the most influential academic chemistry journal, *Reports of the German Chemical Society*, was published in Germany) and trade organizations actively promoted the industry interests (Murmman 2003).

Social movements constitute another aspect of the institutional landscape (Rao and Sivakumar 1998) and exhibit the ability to legitimate new technologies and markets (Carroll and Swaminathan 2000; Dowell et al. 2002; Greve et al. 2006; King and Pearce 2010; Lounsbury et al. 2003; Sine and Lee 2009; Sine et al. 2005; Weber et al. 2009). Rao (2002, 2009) highlights the pivotal role social movements played in legitimizing the automobile in the early 20th century over the prevailing mode of transportation, horse-drawn carriages, and other alternatives like the electric- and steam-powered automobiles. Voluntary associations (i.e. clubs for automobile enthusiasts) that mobilized in support of the automobile were critical for disseminating information and organizing races or “reliability contests,” which received extensive coverage from newspapers and promoted gasoline-powered automobiles as a safe and reliable form of transportation.

The relative view of innovation

Innovation is not only influenced by social network, organizational, and institutional arrangements; whether a product is considered novel depends on one’s point of view (Dahlin 2011). Alternatively situated observers may have different perceptions or cognitive understandings of the same idea or material object. An idea or object may be perceived as novel by one observer while an observer with a different point of view may consider the object completely unoriginal. I refer to this perspective as the relative view of innovation. The subjective nature of innovation is not new to the innovation literature. In his influential monograph on diffusion, Rogers (1995) maintains that an idea or object is novel depending on whether it is perceived as novel. An idea is perceived as novel by someone to whom it was previously unknown (see also Van de Ven and Rogers 1988; Van de Ven et al. 1999).

Research conducted by proponents of actor-network theory (ANT) in studies of science, technology and society (STS) illustrates how one’s point of view influences her perception of material objects.² Actor-network theorists are notable advocates of the view that human and nonhuman “actors” possess agency. Consequently, a human actor’s perception of a nonhuman object is influenced by her relationship to it. This view has two primary implications for the sociology of innovation. First, knowledge-creation is influenced by interactions

between human and nonhuman actors (Callon 1991; Latour 1988, 2005). A study by Beunza and Stark (2004) of analysts in a trading room demonstrates how the relationship between humans and material objects impacts human behavior by fostering innovation. Beunza and Stark describe how the firm periodically altered the configuration of material objects in the trading room, which in turn changed the physical proximity of traders to one another. Rotating the physical location of traders' desks facilitated social interaction and increased knowledge sharing. In another study, Beunza and Stark (2012) illustrate how the use of financial models sensitized analysts to new information that influenced their perceptions about the viability of mergers.

A second, related implication of ANT for innovation is that various perspectives of an object may produce differing narratives about the object. By examining the scientific and medical advances of Louis Pasteur, Latour (1991) illustrates how relationships between individuals and objects shape individual views of those objects. Advancing fermentation methods in the production of wine was considered a worthy endeavor by Pasteur, but other scenarios could be imagined where those who were asked to fund Pasteur's research on fermentation may have viewed it as unprofitable or uninteresting. In another account of Pasteur's "discovery" of microbes, Latour (1988) shows that a microbe was seen as a health risk by some but as beneficial by others. With respect to innovation, it follows that individual accounts "vary as a function of the perspective of the observer" (Latour 1991, p. 127). Perceptions of novelty can be challenged as in the case of the discovery of the structure of DNA. Latour (1987) recounts the contested nature of the discovery of DNA's double helix structure, which was attributed to James Watson and Francis Crick in 1953. However, the renowned chemist Linus Pauling had written a paper earlier in 1953 that (incorrectly) identified DNA's structure (see also Watson 1968). And a couple of years earlier, Rosalind Franklin developed X-ray images of DNA's structure that were central to Watson and Crick's "discovery." According to ANT approaches, accounts of an object's utility, relevance, or novelty, depend on individual as well as collective evaluations of it. Whether someone is persuaded that a material object is novel is inextricably linked to her experience, opinion, and understanding; the opinions expressed by others; the reputation and power of those who express opinions; the knowledge of scientific practices and discourse that may justify one object over other available alternatives; and other socially contested and political processes.

Paths to organization innovation

The relative view of innovation does not imply all vantage points are equal (Latour 1991). Certain perspectives are more insightful or relevant than others. Recall the CEO who diagnosed his firm's drug development efforts as a radical departure from other drugs on the market, but upon further reflection, each additional new drug introduced represented an incremental change compared to the firm's existing products. Although this anecdote illustrates the notion of relative innovation, it turns out that the future of this company is bleak. The drugs it has been developing have not received approval from the US Food and Drug Administration and are not yet producing revenue for the firm. Generating a drug that is novel from the perspective of the environment has not been effective for this firm so far. Perhaps there is good reason other firms have not pursued this new drug technology. A new idea is not necessarily a good idea and building on firm competencies in new product creation does not guarantee success.

Which point of view should be privileged? The corollary question with respect to organizational innovation is: Should organizations pursue a path to innovation that builds on firm competencies or knowledge and resources available in the environment? The notion

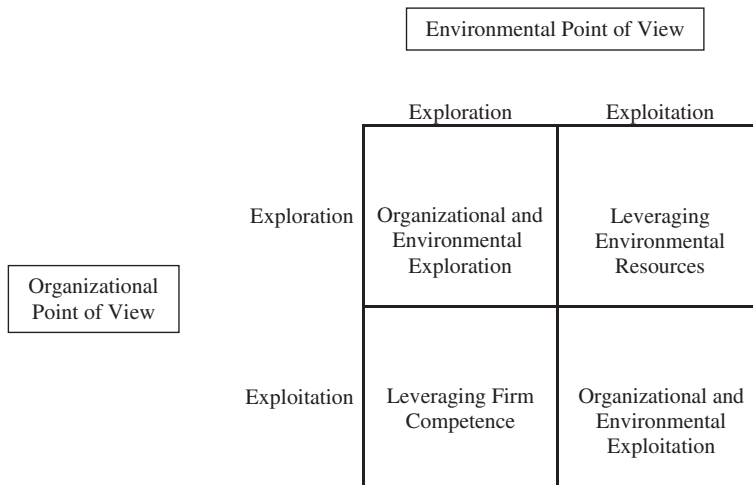


Figure 1 Paths to innovation.

of relative innovation suggests several possible paths based on March's (1991) conceptions of exploration and exploitation. These paths are identified in Figure 1 and include some combination of exploration and exploitation activities with respect to the organization and the environment. Because the novelty of a product can be evaluated with respect to multiple vantage points, the figure contains two dimensions: the organization and the environment. The rows reflect organization exploration or exploitation, while the columns represent environmental exploration or exploitation.

The path identified in the upper left cell is organizational and environmental exploration. A product in this category represents a deviation from an organization's products and is new to the environment. An example is the introduction of the drug thalidomide in the 1950s. Although thalidomide is currently used to treat several medical conditions like AIDS, cancer, and leprosy, thalidomide was originally patented by the German pharmaceutical company Grunenthal and sold on the market in 1957 to treat morning sickness in pregnant women (until it was tragically discovered to cause birth defects). The only drug with which Grunenthal had experience prior to selling thalidomide was penicillin (Grunenthal 2013). Thus, thalidomide represents a product with which the firm Grunenthal had no previous experience, organizational exploration. And Grunenthal was the first to patent and sell the drug on the market so the drug can also be viewed as environmental exploration.

The cell in the upper right corner of Figure 1 is labeled as leveraging environmental resources because it represents an innovation that is new to the organization (organizational exploration) but not the environment (environmental exploitation). During the 1990s, IBM transformed itself from one of the world's leading computer hardware manufacturers to a very successful consulting services and software company. IBM now delivers a broad range of consulting and software solutions to its new customer base. Consequently, IBM's current product offerings could be considered organizational exploration, since its previous experience included computer hardware, and environmental exploitation, since the type of consulting services IBM offers are already familiar to its new clients.

Leveraging firm competence occurs when firms utilize their competencies (organizational exploitation) in ways that are unfamiliar to others in the market (environmental exploration) (lower left cell). This path is illustrated by a product development strategy that is utilized frequently by W. L. Gore & Associates. Gore has made its mark using polytetrafluoroethylene

(PTFE) to modify and improve a wide range of products like high-performance medical implants, gaskets, and seals. The same qualities that make PTFE an effective coating for other materials, weather resistant and strong, are the same properties that made it valuable for clothing that was first developed by Gore in 1978. GORE-TEX® fabric represented “*the first truly waterproof, breathable outerwear*” (italics added) (<https://www.gore-tex.com/remote/Satellite/home>; last accessed 25 March 2014). Accordingly, Gore applied its expertise with PTFE (organizational exploitation) to create a fabric that was new to the clothing and apparel industry (environmental exploration).

The final path to innovation includes organizational and environmental exploitation (lower right cell). Apple, Inc. is especially proficient at leveraging its expertise while appropriating technology from the environment in new product creation. To create the iPhone, Apple incorporated external technology like a digital camera and cell phone into its product. Apple also contracted with FingerWorks, a company that developed the multi-touch interface technology used in the iPhone. When Apple searched for the right type of glass to use for the face of the iPhone, Jobs found gorilla glass, as it is called by its developer, Corning Glass. Gorilla glass, which is incredibly durable, was originally created in the 1960s, but Corning halted its manufacture because a use could not be found for the glass (Isaacson 2011). The 4S version of the iPhone is equipped with another technology from the environment – voice-recognition software known as Siri. Voice-recognition technology has existed for decades in many incarnations. Google produced a voice application for the original iPhone with many of the capabilities possessed by Siri. In the 2000s, voice-recognition capabilities could be found in Windows Vista and Mac OS X, and Android phones include a “Voice Search” function. Apple worked with Siri, Inc., and later acquired the firm in 2010, to develop the Siri technology for use in the 4S. A 2007 *Time* magazine article that was published shortly after the launch of the iPhone proclaimed, “When you get right down to it, the device doesn’t even have that many new features—it’s not like Jobs invented voicemail, or text messaging, or conference calling, or mobile Web browsing” (Grossman 2007).

In addition to exploiting technology from sources external to the firm, Apple leveraged its expertise with other devices to create the iPhone, including experience with MP3 players, web browsers, WiFi capabilities, and operating systems. The iPhone and iPad share many technologies. Both include Internet capability, a camera, a music player, gorilla glass, and a multi-touch interface. Exploitation activities are not limited to technology or product platforms. Expertise includes design, work processes, and other transformational processes used to develop or manufacture a product. Besides exploiting its technological expertise, Apple Inc. utilizes competencies in design, marketing, software, and hardware when it creates new products like desktop computers, tablets, music devices, and smart phones. Apple’s innovation success, therefore, involves combining firm expertise in design, hardware, software, manufacturing, and marketing across product platforms and the ability to integrate technology available in the environment.

This path, organizational and environmental exploitation, leverages knowledge and resources available in both the firm and environment, which in turn can form the basis for an hypothesis about which innovation path is most beneficial to pursue: All things being equal, firms that engage in organizational and environmental exploitation minimize the risks inherent in innovation and increase the likelihood of innovation success. Exploiting existing knowledge and resources reduces uncertainty by helping firms learn from its past and signaling legitimacy to other firms, customers, and constituents in the environment (DiMaggio and Powell 1983; Meyer and Rowan 1977). Generating novelty by recombining existing competencies, knowledge, or resources does not preclude the creation of radically new

alternatives. But as Hargadon and Douglas (2001) point out, to be accepted, entrepreneurs must locate their new products and ideas within preexisting understandings, networks, and institutions that constitute the institutional environment. Hargadon and Douglas go on to explain that successful entrepreneurs draw on preexisting understandings by designing features of new products that invoke familiar schema – mental models or frameworks that guide the interpretation of information – that shape whether customers are likely to understand the relevance of a new product and how to use it.

Fligstein's (1997, 2001; Fligstein and McAdam 2012) concept of 'social skill' provides additional insight into how successful innovators and entrepreneurs motivate the acceptance of new ideas among others. By relating to other people and their circumstances, actors with social skill appeal to shared meanings and persuade others to cooperate. Consider Jacques Delors, president of the European Commission (EU) during the 1980s, who was attempting to consolidate EU economic and political programs and integrate its members. Reform efforts continually received insufficient support, but Delors was able to overcome the opposition by persuading member-states to support one reform at a time. Once states agreed to one program like the Single Market Program, Delors was able to generate support for additional reforms. "The basic idea was to use the commitment to the Single Market as the mechanism to keep proposing new forms of cooperation" (Fligstein 1997, p. 404). Delors justified subsequent expansive reforms like a common currency by invoking the logic of a unified, single market that was already established.

Conclusion

The sociology of innovation in general, and the relative view of innovation in particular, provides insight into the innovation paradox: innovation is necessary for survival, especially in rapidly changing environments, but innovation requires costly investment in new equipment and training and can disrupt existing routines. While ambidexterity scholars suggest that firms overcome the challenges inherent in the innovation paradox by pursuing some combination of exploration and exploitation with respect to the firm, it is also beneficial to consider that a new product can be perceived as new or familiar relative to the environment. An implication of the relative view is that leveraging knowledge and resources from both the firm and environment reduces the risks inherent in the innovation paradox that may prevent innovation success.

Drawing on the relative view of innovation, I develop a typology that identifies multiple paths to innovation. The typology characterizes a new product as exploration or exploitation with respect to the organization and the environment, and the multiple paths consist of some combination of these activities. While it is all but impossible to predict the complexities and contingencies of the market that render some paths more successful than others, it stands to reason that following the innovation path that builds on existing competencies and reduces the uncertainty is likely to decrease the risks associated with innovation. The path that most closely accomplishes this feat includes firm and environmental exploitation, which exploits knowledge and resources available within the firm and from the firm's environment.

Research on innovation demonstrates that building on knowledge, resources, and practices plays a key role in creating enduring solutions (e.g., Fleming 2001; Haveman 1992; Yayavaram and Ahuja 2008). The economist Brian Arthur (2009) elaborates an economic history of technological evolution as a state of perpetual novelty generated by the deconstruction and reconstruction of technological building blocks, and sociologists advocate the notion that social and cultural building blocks can be recombined and directed toward social and political change (Clemens 1993, 1997; Edelman and Stryker 2005; Kelly 2003; Morris 1981; Pedriana and Stryker 1997; Schneiberg 2006; Stark 1996). But, the relative view of

innovation presented in this paper goes a step further. The relative view recasts innovation with respect to multiple reference points, which include the firm and the environment. A new product can exploit the expertise of the firm and resources in the environment concurrently. Hence, successful innovation does not need to depend on the herculean efforts of an individual or collective to reinvent the wheel. Rather, it depends on leveraging existing parts of the wheel and environment to generate new ways to travel, whereby previously impossible expeditions over uncharted terrain become entirely possible.

Short Biography

Eric Dahlin studies institutional and organizational change. His work has been published in *Social Forces*, *Sociological Quarterly*, and *Sociological Inquiry*. Current research projects examine innovation in the pharmaceutical industry and the impact of social norms on the success of engineering projects in developing countries. He is currently an assistant professor in the Department of Sociology at Brigham Young University.

Notes

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¹ To be sure, the concepts of exploration and exploitation represent a simplistic dichotomy. Many times, a new product or service does not represent either exploration or exploitation but contains elements of both. However, these constructs provide a useful orienting framework and illustrate how firms often combine existing knowledge and resources (exploitation) in new ways to generate novel products (exploration) as discussed later in the paper.

² This paper benefitted greatly from feedback by Nicholas Rowland on actor-network theory.

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